

$$(*) (57)_{10} = (\quad)_{BCD}$$

$$\text{Ans: } \underline{01010111}_{BCD}$$

$$(57)_{10} = (01010111)_{BCD}$$

• 1's complement, 2's complement

$$(15)_{10} = (00001111)_2$$

1's complement
2's complement
1's complement
2's complement
1's complement
2's complement

$$(*) 15 + 13 = 28$$

Note:

$$(15)_{10} = (00001111)_2$$

$$(13)_{10} = (00001101)_2$$

1's complement: 11110010

$$2's \text{ complement: } \underline{11110011}_2$$

$$(-13)_{10} = (11110011)_2$$

HW: Find (13-15) using 2's complement.

$$= (2)_{10}$$

$$= (10)_2$$

$$= (00000010)_2$$

$$\begin{array}{r} \text{Carry bit} = 1 \\ 01000001 \end{array}$$

$$11001111 = (-13)_{10}$$

$$11110000 = (-15)_{10}$$

ew

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BCD

$$(*) (57)_{10} = (?)_2$$

$\Rightarrow$

$$(57)_{10} = (111001)_2$$

$$57 = 32 + 16 + 8 + 4 + 1$$
$$1 \quad 1 \quad 1 \quad 1 \quad 1$$

Decimal to Binary

$$0 \longrightarrow 0$$

$$1 \longrightarrow 1$$

$$2 \longrightarrow 10$$

$$3 \longrightarrow 11$$

$$4 \longrightarrow 100$$

$$5 \longrightarrow 101$$

$$6 \longrightarrow 110$$

$$7 \longrightarrow 111$$

$$8 \longrightarrow 1000$$

$$9 \longrightarrow 1001$$

Number System

Decimal: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, ⁺¹10, 11, 12, 13, 14, 15, 16, ...

... 99, 100, 101, 102, 103, ...

999, 1000, 1001, ... (50)

Octal: 0, 1, 2, 3, 4, 5, 6, 7, 10, 11, ⁺¹12, 13, 14, 15, 16, 17, 20, 21, 22, ...

12, 13, 14, 15, 16, 17, 20, 21, 22, ...

23, 24, 25, 26, 27, 30, ... 75, ...

76, 77, 100, 101, ... 777, 1000, ...

1001, ...

Hexadecimal: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F, 10, 11, 12, 13, ...

17, 18, 19, 1A, 1B, 1C, 1D, 1E, 1F, ...

20, 21, ... FE, FF, 100, 101, ...

Binary: 0, 1, 10, 11, 100, 101, 110, 111, 1000, ...